

Machine Learning for Natural Language Processing

Introduction to Transformers



UNIVERSITY OF
GOTHENBURG

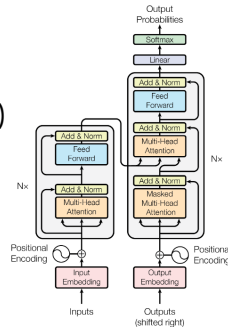
CHALMERS

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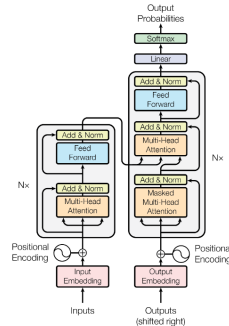
the Transformer

- the **Transformer** architecture (Vaswani et al., 2017) now dominates NLP research
- it is a **family** of architectures!



the Transformer: main idea

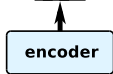
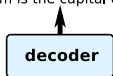
- the Transformer uses **attention** for information flow: *“Attention is all you need”*
- it was developed to mitigate drawbacks of RNNs
- it was originally designed for **machine translation**, and the original design included two modules:
 - an **encoder** that “summarizes” an input sentence
 - a **decoder** that generates an output



types of Transformers: high level

encoder/decoder

Stockholm is the capital of Sweden.



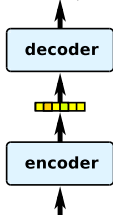
Stockholm är Sveriges huvudstad.

models $P(Y|X)$

types of Transformers: high level

encoder/decoder

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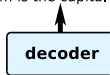


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models $P(Y|X)$

decoder only

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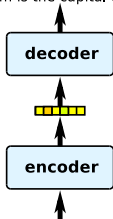


models $P(Y)$

types of Transformers: high level

encoder/decoder

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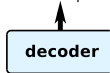


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models $P(Y|X)$

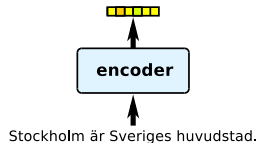
decoder only

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models $P(Y)$

encoder only



outputs a
representation $f(X)$

illustration of a Transformer encoder block

Jane lives in Paris . She is a dentist .

illustration of a Transformer encoder block

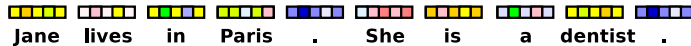


illustration of a Transformer encoder block

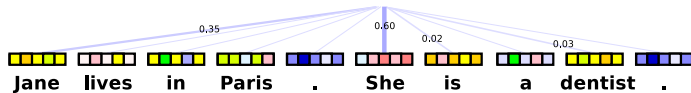


illustration of a Transformer encoder block

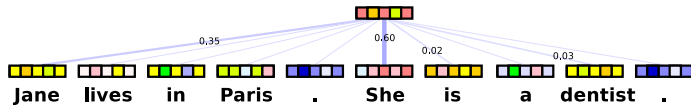


illustration of a Transformer encoder block

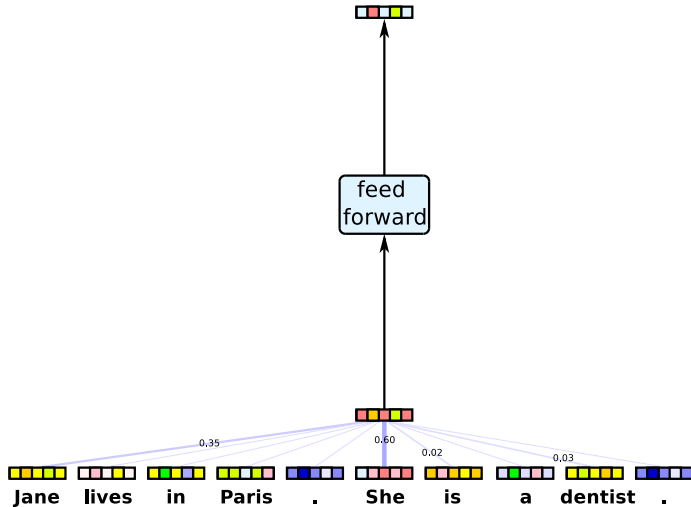
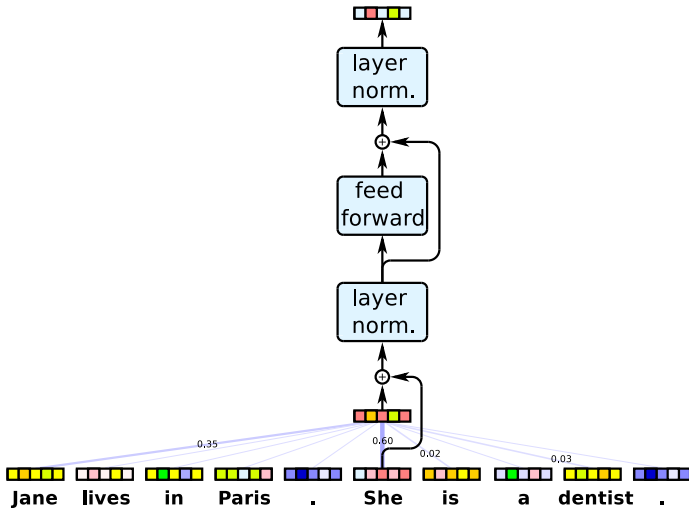


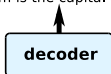
illustration of a Transformer encoder block



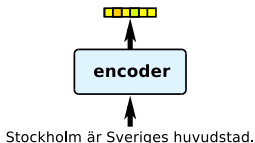
next

Transformer language models: implementation, applications

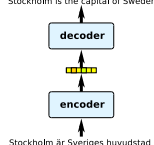
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Later: representation models, translation models



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references

- J. Ba, J. Kiros, and G. Hinton. 2016. [Layer normalization](#). arXiv:1607.06450.
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. 2017. [Attention is all you need](#). In *NIPS* 30.